

The Plant Innovation Centre (PIC) is a world-class research, development, and extension (RD&E) facility. By providing innovative solutions that detect, identify and manage plant pests, our team helps strengthen Australia's biosecurity.

The PIC team have significant scientific expertise and experience in areas such as:

- plant pathology
- diagnostics
- molecular biology
- microbiology
- biotechnology.

The Plant Innovation Centre supports Honours and higher-degree (MSc/PhD) students with generous scholarships, support for travel and development, opportunities to work in state-of-the-art facilities and build networks with a highly sought-after employer.

Get involved

The PIC team are a highly motivated group of scientists who care about protecting our environment and agricultural industries from exotic pests and diseases. If you're passionate about science, [get in touch with us today](#).

[Find out more](#) about our scholarships for honours, masters and PhD students.

We also offer placements for undergraduates via the University Vacation Employment Program (UVEP).

Current projects

Faster seed testing and release: MicroRNA biomarkers for seed infection

A jointly funded project with CSIRO under the Catalysing Australia's Biosecurity to investigate the potential of RNA biomarkers as a rapid, simple test for the infection status of seeds.

Development of emerging pathogen detection tool, linking published detections, sequences and reports (Seq-er)

Development of an automated, bioinformatic tool to find, extract and collate information on exotic plant pest detections from publicly available international databases.

Sustained storage of viable virus-infected plant positives at PEQ glasshouses

An investigation into safer and longer-term storage options of positive plant material used for testing and innovation purposes.

Harmonisation of PCR cycling to improve diagnostic workflow efficiencies

This project will optimise a set of standard cycling conditions to reduce the detection time on key fungal pathogens.

Develop and implement an agreed 'emergency' validation method for new assays

This project aims to develop a validation template to fast track newly developed assays in an emergency context.

Developing rapid, accurate and field deployable diagnostics (CRISPR-Cas)

CRISPR is a gene-editing tool that we have demonstrated can be used to detect *Xylella* (Australia's #1 priority plant pest). We are refining this technology to use it in the field and target multiple pests in a single tube.

Investigating alternate treatments to devitalise seeds

A partnership project with Steritech to investigate the potential of electron beam (E-beam) irradiation to treat commodities such as seeds, grains, and cut-flowers. Using e-beam instead of other treatments (such as chemicals, heat or steam) will result in faster, cheaper, safer imports.

Rapid detection of plant disease by microneedle patch array

A partnership project with the University of Southern Queensland to trial microneedles as a rapid and reliable method to detect plant pests. If successful, microneedles could be fused to CRISPR and/or lab-on-a-chip technology to improve our ability to accurately and quickly detect dangerous pests and pathogens.

Development of a cost-effective method using third-generation sequencing to detect pests in-planta

PhD project collaboration with the University of the Sunshine Coast to use MinION technology to rapidly detect and identify exotic viruses in imported seeds. This technology will make seed imports cheaper and safer.

Plant disease diagnostics in tissue culture

PhD project collaboration with the Plant Biosecurity Training Centre and James Cook University aiming to develop a reliable plant disease assay that is directly applicable to tissue culture material.

Improving TEM microscopy

The department routinely uses transmission electron microscopy to visualise virus particles; this project will review current techniques and update our methods to align with international best-practice.

Seed-borne pest assays

The PIC team is developing several novel qPCR assays to reliably detect exotic seed-borne pests.

iMDAC-DC, in silico Molecular Diagnostic Assay Components Design and Check

This project uses AI and machine learning to develop a bioinformatic tool to streamline molecular assay development for regulatory diagnostics. Currently this process is time and labour intensive, creating a bottleneck in assay design and review. A partnership with NSW DPIRD and Deakin University.

Whole genome sequencing of the Australian Plague Locust

The Australian Plague Locust (APL) is a major agricultural and environmental pest. Typically controlled by chemicals, new approaches are needed due to limitations on chemical use. Sequencing the genome of the APL will provide valuable insights into potential alternative management strategies.

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Novel disinfestation strategies as alternatives to methyl bromide (MB)

A partnership project with the Victorian Strawberry Industry Certification Authority to trial methyl iodide as an alternative to MB as a fumigant treatment. MB is a potent greenhouse depleting gas and banned in many countries. Due to the success of this project, it has been expanded and funded by [Forest and Wood Products](#).

Improved diagnostics for exotic fungal pathogens affecting commercial mushroom production

A partnership project with Agriculture Victoria to develop molecular diagnostic tools to detect and identify mushroom pathogens. These pathogens currently take weeks to detect; molecular tools will achieve this in hours.

Lab-on-a-chip assay for the rapid detection of multiple plant viruses

A partnership project with RMIT University to trial the use of nanotechnology to detect priority plant pests. If successful, this technology could be adapted as a cutting-edge, highly-portable diagnostic tool allowing faster detections and reducing our reliance on expensive laboratories.

Improved molecular diagnostics to detect exotic bee mites

A partnership project between PIC and CSIRO aims to improve our ability to detect exotic mites in honeybees using molecular diagnostic tools. This will improve our ability to rapidly respond to detections of exotic mites.

Alternative glyphosate herbicides

We trialled dozens of alternative herbicides to replace glyphosate as a treatment for imported cut-flowers. Glyphosate alternatives will give exporting countries greater options to help facilitate trade.

Improve Post Entry Quarantine diagnostics using High throughput sequencing (Hts)

Working with government, industry, and university partners we successfully implemented Hts for testing high-risk plants in December 2022. Hts replaces hundreds of individual tests with a single test to detect all viruses at once. This results in cheaper, faster, safer plant imports.

Rapid identification of unknown insects at the border using MinION sequencing

Using nanopore sequencing, the PIC team developed new methods to identify unknown insects in hours, compared to days. This means goods can be cleared and released from biosecurity control much faster.

Rapid identification of unknown seeds at the border using MinION sequencing

Following on from our success using MinION, to rapidly identify unknown insects, we applied this technology to identify unknown seeds. Although technologically feasible, more database development is required before implementation.

Development of new tests for begomoviruses

Begomoviruses are responsible for considerable economic damage to many important agricultural crops. We developed diagnostic tests to more accurately detect newly emerging begomovirus risks.

Verification of imported cut flowers using HPLC-Mass spec

Imported cut flowers are treated offshore to manage biosecurity risk. We developed and evaluated a new test to verify that treatments were being conducted as required.

Evolution of Plant Virus Diagnostics used in Australian PEQ '[Plants](#)' July 2021

A Rapid and Cost-Effective Identification of Invertebrate Pests at the Borders Using MinION Sequencing of DNA Barcodes '[Genes](#)' July 2021

Complete genome sequence of a novel potyvirus infecting *Miscanthus sinensis* '[Archives of Virology](#)' November 2021

Side-by-Side Comparison of Post-Entry Quarantine and High Throughput Sequencing Methods for Virus and Viroid Diagnosis '[Biology](#)' February 2022

Implementation of GA-VirReport; A Web-based Bioinformatics Toolkit for Post-Entry Quarantine screening of Virus and Viroids in Plants '[Viruses](#)' July 2022

Development of a rapid, accurate, and field deployable LAMP-CRISPR-Cas12a integrated assay for *Xylella fastidiosa* detection and surveillance '[Australasian Plant Pathology](#)' November 2023

First detection and complete genome sequence of a new potexvirus naturally infecting *Adenium obesum* '[Archives of Virology](#)' September 2023

A review of probe-based enrichment methods to inform plant virus diagnostics '[International Journal of Molecular Sciences](#)' July 2024

Managing regulatory issues arising from new diagnostic technologies: A case-study using high throughput sequencing '[CABI Agriculture and Bioscience](#)' March 2025

Use of Multi-Locus Metabarcoding to Inform an Australian Government Biosecurity Response on the Origins of Suspected Illegal Plant Products '[International Journal of Molecular Sciences](#)' June 2025

A rapid and sensitive RPA-Cas12a assay for detection of banana bunchy top virus (Babuvirus musae) '[Tropical Plant Pathology](#)' June 2026

Polymeric microneedle patches for rapid DNA extraction and molecular diagnosis of foliar pathogens of mung bean '[Talanta](#)' July 2026

PROJECT PARTNERS

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